

Table 3: Top 10 Candidate Anode Host Materials by Multi-Criteria Composite Score

formula	anode_composite_score	E_ads_DFT_eV	E_ads_DFT_eV_pred	band_gap	band_gap_pred	energy_hull	energy_hull_pred	bulk_modulus	bulk_modulus_pred	pugh_ratio
C	0.765	-2.830	-2.821	3.536	3.618	0.242	0.555	419.280	410.401	0.882
FeOF	0.696	-2.047	-2.088	0.000	0.005	1.111	0.895	172.310	185.617	1.905
NbF3	0.692	-3.058	-2.582	0.000	0.014	0.587	0.724	95.030	95.008	2.114
HgF2	0.655	-2.734	-2.669	0.862	1.326	0.000	0.000	97.630	114.448	5.002
Mn2F7	0.649	-2.554	-2.337	0.000	0.000	0.749	0.734	43.040	42.421	1.846
Mg2SiO4	0.643	-2.847	-2.516	4.021	3.961	1.423	1.295	193.240	188.701	1.655
SmZnPO	0.640	-3.081	-2.778	1.474	1.823	0.000	0.216	94.000	101.569	1.709
VOF	0.639	-2.935	-2.814	0.633	1.069	1.099	1.252	91.080	98.122	1.907
TiTi2F6	0.633	-2.950	-2.638	3.589	3.709	0.000	0.003	37.110	58.339	2.695
Rb2F	0.622	-2.035	-1.900	0.000	0.000	0.017	0.000	10.710	26.206	2.385